Fast Orthogonal Projection for Hyperspectral Unmixing

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***Abstract*— Spectral unmixing plays a vital role in hyperspectral image analysis. It mainly consists of two procedures, i.e., endmember extraction and abundance estimation. Although most algorithms for each of the two procedures may exhibit good performance, few studies have been done considering both problems simultaneously. Therefore, hyperspectral unmixing accuracy is normally achieved by exploring all possible combinations of the two types of algorithms, which renders high computational overloads. We propose a novel orthogonal projection framework to conduct fast hyperspectral unmixing. It addresses both endmember extraction and abundance estimation with orthogonal projection endmember (OPE) and orthogonal projection abundance (OPA). Especially, the pixel with the largest orthogonal projection on any pixel is considered to be an endmember. We randomly choose one pixel from the hyperspectral data to compute the orthogonal projections of all pixels and extract the pixel with the largest projection as the first endmember. To avoid extracting the same endmembers, we compute orthogonal projections of all pixels to endmembers that have been previously extracted, and the pixel with the largest projection is considered as the next endmember. In terms of abundance estimation, we also utilize the concept of orthogonal projection and search for a diagonal matrix whose multiplication with the endmember matrix is not only a square matrix but also a diagonal matrix. Then, we exploit some specific matrix operations to estimate the abundance of each endmember at every pixel. We have evaluated the proposed OPE and OPA algorithms on synthetic and real data, and the experimental results have validated their effectiveness and efficiency in hyperspectral unmixing.**

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I. Introduction

HYPERSPECTRAL images (HSIs) contain detailed and rich spectral and spatial information, which has become the key to identifying the composition of ground scenes and has been applied to numerous fields [1]–[5]. A typical problem associated with HSIs is the low spatial resolution of sensors, which inevitably leads to the existence of mixed pixels. Especially, hyperspectral unmixing [6]–[10] is exploited as an outstanding tool to address the problem by decomposing mixed pixels into a set of pure spectra (i.e., endmembers) and their corresponding proportions (i.e., abundances). Over the past decades, many hyperspectral unmixing techniques have been developed in the literature. These techniques are divided into three main groups, i.e., determining the number of endmembers, extracting their spectral signatures, and estimating the corresponding abundances. Generally, most endmember extraction algorithms assume that the number of endmembers has been known in advance through other strategies. Therefore, in our work, we mainly focus on the two tasks of endmember extraction and abundance estimation. In Sections I-A and I-B, we review the literature about endmember extraction and abundance estimation, respectively. In Section I-C, we briefly describe the major contributions of our work in terms of presenting a new hyperspectral unmixing framework that simultaneously extracts endmembers and estimates the abundance of each endmember at every pixel.

A. Endmember Extraction

The related algorithms for endmember extraction are mainly divided into two groups according to whether there are pure pixels in the HSI or not. Some typical algorithms implemented using the pure pixel assumption include orthogonal subspace projection (OSP) [11], vertex component analysis (VCA) [12], and N-FINDR [6], [13], among others. OSP first selects the pixel vector with the largest length in the scene as the first endmember and then finds the pixel vector with the maximum absolute projection as the second endmember in the space orthogonal to the space linearly spanned by the initial pixel. The third endmember is extracted by applying an orthogonal subspace projector to the original image, where the signature

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that has the maximum orthogonal projection in the space orthogonal to the space linearly spanned by the first two endmembers. OSP repeats the process until all endmembers are found. Both VCA and N-FINDR are devised to search for a simplex with the maximum volume, and endmembers are the vertices of the simplex. VCA iteratively projects data onto an orthogonal direction to the subspace spanned by the endmembers already extracted, and the pixel with the extreme of the projection is the next endmember. N-FINDR randomly chooses some pixels to form an initial endmember matrix and then replaces the pixels with other pixels to find a set of pixels with the largest volume to extract endmembers. To avoid identifying the same endmember, an automatic target generation process (ATGP) [14] is proposed, and it utilizes the notion of OSP to extract endmembers. To address the impact of noise for endmember extraction, alternating decoupled volume max-min (ADVMM) and successive decoupled volume max-min (SDVMM) [15] are developed for dealing with the worst case simplex volume maximization problem by alternating optimization and successive optimization, respectively. Moreover, considering that spectral variability plays a vital role in spectral unmixing, Drumetz *et al.* [16] proposed to reinterpret the concept of convex geometry used in spectral unmixing in the cases of illumination induced and intrinsic variability in a linear mixing model (LMM) framework and discussed the validity and extension of concepts, such as endmember extraction, intrinsic dimensionality, and pixel purity. In addition, based on the insights gained from this analysis, Drumetz *et al.* [16] proposed an integrated unmixing chain for this situation, which is still based on the concept of convex geometry but adapts it to the new geometry introduced by the presence of variable endmembers. Most unmixing methods require the number of endmembers and have high computational overloads. To address these problems, Das *et al.* [17] proposed two new frameworks for spectral unmixing under the known and unknown number of endmembers, respectively. Das *et al.* [18] presented a new and fast convex geometric framework for spectral unmixing. The new framework detects the extreme points to extract endmember signatures.

If there are no pure pixels in HSIs, endmember extraction will become a very challenging task. The difficulty lies in the fact that HSIs may not contain any endmembers or at least some of them. Some popular algorithms implemented without the pure pixel assumption include minimum-volume enclosing simplex (MVES) [19], minimum-volume simplex analysis (MVSA) [20], [21], robust MVSA (RMVSA) [22], polytope decomposition (POD) [23], and so on. Among representative algorithms in this category, MVES incorporates convex analysis and Craig's criterion to extract endmembers, and it searches for a simplex containing all the dimensionally reduced pixels by minimizing the simplex volume subject to the constraint. MVSA fits a minimum-volume simplex to HSIs by constraining the abundance fractions to belong to the probability simplex. RMVSA is a version of MVSA that is robust to noise, and it develops the relaxation of nonlinear chance constraint, which greatly reduces the computational complexity of nonlinear chance constraint problems. Especially, RMVSA can be used in any unmixing algorithm that uses nonlinear chance constraints. POD provides an effective method to get the abundance in a nonlinear mixture, which can be approximated by a polynomial model and is independent of the order of the nonlinearity.

In the above two cases, it is a great challenge to directly identify realistic endmember signatures from the HSIs without prior knowledge. In this scenario, some methods exploit endmember signatures from a previously available spectral library, such as the U.S. Geological Survey (USGS),[1] which is potentially very large. Then, a best subset of the endmembers of the library will be chosen to model each pixel [24]. Tang *et al.* [25] developed a new unmixing method called sparse unmixing using spectral *a priori* information (SUnSPI), which uses the spectral *a priori* information in the HSI to alleviate the problem of high mutual coherence of the spectral library faced by sparse unmixing. Shi *et al.* [26] presented a novel algorithm termed collaborative sparse hyperspectral unmixing using l_0-norm (CSUnL0) for the sparse hyperspectral unmixing problem to address the NP-hard problem caused by the l_0-norm constraint. Xu and Shi [27] transformed sparse unmixing to a biobjective optimization problem to present a new multiobjective-based sparse unmixing method, which addresses the sparse unmixing problem without any relaxation. Das *et al.* [28] employed a sparsity measure for dictionary pruning and presented a new sparsity-based dictionary-aided unmixing framework, which is nonparametric and noise robust with satisfactory performance on the spectral library with high mutual coherence. However, endmember signatures from spectral libraries usually present inconsistent acquisition conditions compared to the image pixel signatures, and this fact may affect the result of abundance estimation, which is the subsequent step of endmember extraction.

B. Abundance Estimation

The main models for abundance estimation include the LMM or nonlinear mixing model (NLMM) [29]–[32]. LMM focuses on the assumption that incident light interacts with a single component, and the spectrum of each pixel is formulated as a linear combination of endmembers and corresponding abundances. NLMM works on the actual mechanism of hyperspectral sensors, although it has high computational complexity and requires prior knowledge of the scene. The most popular algorithm for abundance estimation is the fully constrained least squares (FCLS) [33]. FCLS imposes the abundance nonnegative constraint (ANC) and abundance sum-to-one constraint (ASC) and minimizes the square error in the linear approximation of HSIs. In [34], FCLS has been exploited by introducing a variable splitting and solving the optimization problem with the alternating direction method of multipliers (ADMM), denoted as SUnSAL. In [35] and [36], perturbed linear mixing model (PLMM) and augmented linear mixing model (ALMM) approaches are proposed to estimate abundances when the HSI ensures spectral variability.

[1] http://speclab.cr.usgs.gov/spectral-lib.html

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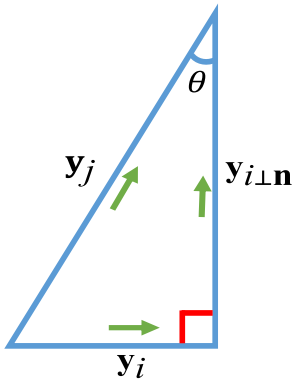


Fig. 1. Schematic representation about the computation of the orthogonal projection of $\mathbf{y}_j$ to $\mathbf{y}_i$, where $\mathbf{y}_i$ and $\mathbf{y}_j$ are two HSI pixels, $\mathbf{y}_{i\perp\mathbf{n}}$ is the orthogonal projection vector of $\mathbf{y}_j$ to $\mathbf{y}_i$, and the green arrow indicates the direction of pixels.

Recently, some researchers consider hyperspectral unmixing as a blind source separation (BSS) problem and bring the application of nonnegative matrix factorization (NMF). NMF-based unmixing not only provides fully unsupervised without the pure pixel assumption but also simultaneously extracts endmembers and estimates their abundances. However, the standard NMF model is nonconvex and shows scale indeterminacy. In this regard, it is not enough to obtain a stable solution by only considering ANC. Therefore, a series of additional constraints [37] is introduced into the standard NMF model. Taking into account the nonconvexity and computational complexity of the simplex volume and the fact that the smallest-volume simplex is equal to the compact simplex possible, MDCNMF [38] has been proposed. MDCNMF exploits the distances between endmembers to measure compactness. Concerning the constraints on the endmembers, minimum-volume constrained nonnegative matrix factorization (MVCNMF) [39] has been introduced. MVCNMF adopts nonnegative matrix factorization and a volume-based constraint to extract endmembers and estimate the abundance of each endmember at every pixel.

The aforementioned algorithms present good performance in the specific task of the unmixing process. However, they also show some drawbacks in computational complexity. Moreover, most hyperspectral unmixing techniques need to combine an endmember extraction algorithm and an abundance estimation algorithm to complete the overall process of unmixing. Therefore, the accuracy of unmixing relies on the combination of the two types of algorithms. However, the fact is that this strategy results in high computation burden by exploiting different combinations of endmember extraction and abundance estimation algorithms to obtain the best performance in hyperspectral unmixing.

C. Contributions

As reviewed in Sections I-A and I-B, hyperspectral unmixing algorithms generally consider endmember extraction and abundance estimation as two independent processes, obtaining good performance in their respective fields, and then, they search for a good combination of the two processes of algorithms to yield the best performance. Another strategy is the NMF-based, which simultaneously completes the two tasks, i.e., endmember extraction and abundance estimation. Whether seeking for the best combination of the two tasks or directly completing unmixing, the two types of hyperspectral unmixing techniques both bring render considerable computation burden, making them inefficient for tackling large-size remote sensing images. To address the problem, we establish a unified and fast orthogonal projection framework for hyperspectral unmixing. To evaluate our proposed framework, we refer to the algorithms of endmember extraction and abundance estimation as orthogonal projection endmember (OPE) and orthogonal projection abundance (OPA), respectively. The contributions of our work are as follows.

Algorithm 1 Pseudocode of OPE for Endmember Extraction

Input:
HSI data set: $\mathbf{Y} \in \mathbb{R}^{L\times N}$
The number of endmembers: c
I = []
$\mathbf{M} = \mathbf{Y}(:, 1)$ {The randomly selected pixel}
for i = 1 to c **do**
  $\mathbf{M}^T\mathbf{n} = \mathbf{0}$
  $\mathbf{n}_{\text{unit}} = \frac{\mathbf{n}}{||\mathbf{n}||_2}$
  for j = 1 to N **do**
    $\hat{I} = \text{argmax}|\mathbf{n}\cdot\mathbf{y}_j|$
  end for
  I = [I, $\hat{I}$]
  $\mathbf{M}$ = [$\mathbf{Y}$(:,I)]
end for
Output:
Endmember Matrix: $\mathbf{M} \in \mathbb{R}^{L\times c}$

Algorithm 2 Pseudocode of OPA for Abundance Estimation

Input:
HSI data set: $\mathbf{Y} \in \mathbb{R}^{L\times N}$
Endmember Matrix: $\mathbf{M} \in \mathbb{R}^{L\times c}$
The number of endmembers: c
The number of pixels: N
$\mathbf{P}$ = []
for i = 1 to c **do**
  $\mathbf{M}(:, i) = []$
  $\mathbf{M}^T\mathbf{p} = 0$
  $\mathbf{P}$ = [$\mathbf{P}$,$\mathbf{p}$];
end for
$\mathbf{Q} = \mathbf{P}^T\mathbf{M}$
$\mathbf{A} = \mathbf{Q}^{-1}\mathbf{P}^T\mathbf{Y}$ {Eq. (22)}
$\mathbf{A} = |\mathbf{A}|$
{Abundance nonnegative constraint (ANC)}
for j = 1 to N **do**
  $\mathbf{A}(:, j) = \mathbf{A}(:, j)/(\sum_{k=1}^{c}\mathbf{A}(k, j))$
  {Abundance sum-to-one constraint (ASC)}
end for
Output:
Abundance Matrix: $\mathbf{A} \in \mathbb{R}^{c\times N}$

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1) We establish a novel orthogonal projection framework for hyperspectral unmixing, which can simultaneously complete endmember extraction and abundance estimation, and it exhibits good effectiveness and high efficiency in hyperspectral unmixing.
2) The proposed OPE algorithm extracts endmembers by utilizing the concept of the pixel with the maximum orthogonal projection on any pixel is an endmember. In particular, to avoid repeatedly extracting the same endmembers, our proposed OPE algorithm exploits the normal vector of all endmembers that have been extracted to compute the projections of all pixels to it and then chooses the pixel with the maximum projection as an endmember. Since it only involves straightforward vector-based computations, its computational complexity is very low. Especially, the high efficiency of our proposed OPE algorithm has been validated in the experiments.
3) Endmembers and their proportions at every pixel are generally considered to be linearly mixed. However, we cannot estimate abundances by computing the multiplication of the data matrix and the inverse of endmember matrix since the endmember matrix is usually not a square matrix. To this end, our proposed OPA algorithm exploits a diagonal matrix, and its multiplication with the endmember matrix is not only a square matrix but also a diagonal matrix. We then obtain the abundance matrix by computing the multiplication of matrices. In this scenario, our proposed OPA algorithm has high efficiency in terms of solving the abundance of endmembers at every pixel. Especially, experimental results have validated that our proposed OPA algorithm has high efficiency in abundance estimation.

D. Article Outline

The rest of this article is organized as follows. In Section II, we detail the proposed OPE algorithm for endmember extraction and the proposed OPA algorithm for abundance estimation. In Section III, we empirically evaluate the performance of the proposed OPE and OPA algorithms in endmember extraction and abundance estimation on synthetic data and Cuprite data. In Section IV, we conclude this article with some remarks and hints at plausible future research lines.

II. Methodology

Let $\mathbf{Y} = [\mathbf{y}_1, \mathbf{y}_2, \ldots, \mathbf{y}_i, \ldots, \mathbf{y}_N] \in \mathbb{R}^{L\times N}$ denote the HSI data matrix with L bands and N pixels, where $\mathbf{y}_i = [y_{1i}, x_{2i}, \ldots, y_{Li}]^T \in \mathbb{R}^{L\times 1}$ denotes a pixel represented by an L-dimensional vector. As shown in Fig. 1, $\mathbf{y}_i$ and $\mathbf{y}_j$ are two pixels of HSI and $\mathbf{y}_{i\perp\mathbf{n}}$ is the orthogonal projection vector of $\mathbf{y}_j$ to $\mathbf{y}_i$. We have

$$\cos\theta = \frac{\mathbf{y}_{i\perp\mathbf{n}}\cdot\mathbf{y}_j}{||\mathbf{y}_{i\perp\mathbf{n}}||_2||\mathbf{y}_j||_2} = \frac{||\mathbf{y}_{i\perp\mathbf{n}}||_2}{||\mathbf{y}_j||_2} \tag{1}$$

where $||\cdot||_2$ represents the L_2-norm. We obtain the orthogonal projection of $\mathbf{y}_j$ to $\mathbf{y}_i$ by

$$\begin{aligned} ||\mathbf{y}_{i\perp\mathbf{n}}||_2 &= ||\mathbf{y}_j||_2|\cos\theta| \\ &= ||\mathbf{y}_j||_2|\frac{\mathbf{y}_{i\perp\mathbf{n}}\cdot\mathbf{y}_j}{||\mathbf{y}_{i\perp\mathbf{n}}||_2||\mathbf{y}_j||_2}| \\ &= |\frac{\mathbf{y}_{i\perp\mathbf{n}}\cdot\mathbf{y}_j}{||\mathbf{y}_{i\perp\mathbf{n}}||_2}|. \end{aligned} \tag{2}$$

Assuming that $\mathbf{n}$ is the normal vector of $\mathbf{y}_i$ and $\mathbf{n}_{\text{unit}}$ is the unit vector of $\mathbf{n}$, (2) is transformed into

$$||\mathbf{y}_{i\perp\mathbf{n}}||_2 = |\mathbf{n}_{\text{unit}}\cdot\mathbf{y}_j| \tag{3}$$

and $|\cdot|$ denotes the absolute value.

A. Endmember Extraction

We use some techniques to filter out the noise in advance. From a geometric point of view, we know that the pixel with the largest orthogonal projection is an endmember. We randomly select one pixel $\mathbf{y}_i$ from HSI dataset and obtain its normal vector $\mathbf{n}^1 \in \mathbb{R}^{L\times 1}$ by addressing

$$\mathbf{y}_i\cdot\mathbf{n}^1 = 0. \tag{4}$$

Especially, it will have many solutions if we solve the normal vector according to (4). In our work, we sum all the solutions of the normal vector as its real solution. Next, we obtain the unit vector $\mathbf{n}^1_{\text{unit}}$ of $\mathbf{n}^1$ by

$$\mathbf{n}^1_{\text{unit}} = \frac{\mathbf{n}^1}{||\mathbf{n}^1||_2}. \tag{5}$$

We utilize (3) to compute the orthogonal projections of all pixels of HSI to $\mathbf{y}_i$ and obtain the index $\hat{I}_1$ of the first endmember by

$$\hat{I}_1 = \underset{I_1\in\{1,2,\ldots,N\}}{\arg\max} \left|\mathbf{n}^1_{\text{unit}}\cdot\mathbf{y}_{I_1}\right| \tag{6}$$

and we then use the index $\hat{I}_1$ to extract the first endmember $\mathbf{m}_1$ from the HSI dataset. At this moment, the endmember matrix is $\mathbf{M}_1 = [\mathbf{m}_1] \in \mathbb{R}^{L\times 1}$. Next, we get the normal vector $\mathbf{n}^2 \in \mathbb{R}^{L\times 1}$ of $\mathbf{M}_1$ by

$$\mathbf{M}_1^T\mathbf{n}^2 = 0 \tag{7}$$

and the symbol T denotes transpose. The unit vector $\mathbf{n}^2_{\text{unit}}$ of $\mathbf{n}^2$ is computed by

$$\mathbf{n}^2_{\text{unit}} = \frac{\mathbf{n}^2}{||\mathbf{n}^2||_2}. \tag{8}$$

We obtain the index $\hat{I}_2$ of the second endmember by

$$\hat{I}_2 = \underset{I_2\in\{1,2,\ldots,N\}}{\arg\max} \left|\mathbf{n}^2_{\text{unit}}\cdot\mathbf{y}_{I_2}\right|. \tag{9}$$

Similarly, we use the index $\hat{I}_2$ to extract the second endmember $\mathbf{m}_2$ from the HSI dataset. Now, the endmember matrix is updated to $\mathbf{M}_2 = [\mathbf{m}_1, \mathbf{m}_2] \in \mathbb{R}^{L\times 2}$. Especially, the reason for constantly updating the endmember matrix is that the projections of all endmembers to their orthogonal vector are zero. Therefore, we compute the orthogonal vector of the new endmember matrix to avoid extracting the same endmembers and reduce the computational complexity of extracting endmembers. We extract the remaining endmember according to the method of extracting the second endmember, and the endmember matrix is updated to $\mathbf{M}_{k-1} = [\mathbf{m}_1, \ldots, \mathbf{m}_{k-1}] \in \mathbb{R}^{L\times(k-1)}$,

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and $\mathbf{M}_{k-1}$ denotes that we have extracted $k-1$ endmembers. To extract the kth endmember, we get the normal vector $\mathbf{n}_k$ of $\mathbf{M}_{k-1}$ by

$$\mathbf{M}_{k-1}^T \mathbf{n}^k = 0 \tag{10}$$

and the unit vector $\mathbf{n}_{\text{unit}}^k$ of $\mathbf{n}^k$ is computed by

$$\mathbf{n}_{\text{unit}}^k = \frac{\mathbf{n}^k}{||\mathbf{n}^k||_2}. \tag{11}$$

We get the index I_k of the kth endmember by

$$\hat{I}_k = \underset{I_k \in \{1,2,\ldots,N\}}{\arg\max} \left| \mathbf{n}_{\text{unit}}^k \cdot \mathbf{y}_{I_k} \right|. \tag{12}$$

We use the index $\hat{I}_k$ to extract the kth endmember and update the endmember matrix to $\mathbf{M}_k = [\mathbf{m}_1, \ldots, \mathbf{m}_{k-1}, \mathbf{m}_k] \in \mathbb{R}^{L \times k}$. We repeat the above process until all endmembers are extracted. We refer to the method of extracting endmembers as OPE. The detailed process of the OPE algorithm is shown in Algorithm 1. From Algorithm 1, we see that our proposed OPE algorithm only involves matrix operations, and its computational complexity is very low. Especially, in Section III, the experimental results also validate that the proposed OPE algorithm exhibits high efficiency in the task of extracting endmembers on synthetic data and Cuprite data.

B. Abundance Estimation

Let $\mathbf{A} = [\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_i, \ldots, \mathbf{a}_N] \in \mathbb{R}^{c \times N}$ represent abundance matrix and c is the number of endmembers. Under the LMM, the data matrix is represented by the multiplication of the endmember matrix and the abundance matrix by

$$\mathbf{Y} = \mathbf{M}\mathbf{A}. \tag{13}$$

From a traditional view, abundance matrix $\mathbf{A}$ is computed by

$$\mathbf{A} = \mathbf{M}^{-1}\mathbf{Y} \tag{14}$$

and $\mathbf{M}^{-1}$ is the inverse matrix of $\mathbf{M}$. Broadly speaking, it is unrealistic to computing abundance matrix $\mathbf{A}$ by (14) because the endmember matrix $\mathbf{M}$ is usually not a square matrix. In this scenario, we find a diagonal matrix $\mathbf{Q}$ ($\mathbf{P}^T\mathbf{M} = \mathbf{Q}$) to complete the task of abundance estimation. Next, we first introduce how to get the matrix $\mathbf{P} = [\mathbf{p}_1, \mathbf{p}_2, \ldots, \mathbf{p}_i, \ldots, \mathbf{p}_c] \in \mathbb{R}^{L \times c}$. Especially, we obtain $\mathbf{p}_1$ by

$$[\mathbf{m}_2, \mathbf{m}_3, \ldots, \mathbf{m}_c]^T \mathbf{p}_1 = \mathbf{0}. \tag{15}$$

In this sense, $\mathbf{p}_1$ is not unique. We sum all the solutions obtained by (15) as the final $\mathbf{p}_1$. We get $\mathbf{p}_2$ by addressing

$$[\mathbf{m}_1, \mathbf{m}_3, \ldots, \mathbf{m}_c]^T \mathbf{p}_2 = \mathbf{0}. \tag{16}$$

We also sum all the solutions obtained by (16) to obtain the final $\mathbf{p}_2$. Similarly, we acquire $\mathbf{p}_i$ by

$$[\mathbf{m}_1, \mathbf{m}_2, \ldots, \mathbf{m}_{i-1}, \mathbf{m}_{i+1}, \ldots, \mathbf{m}_c]^T \mathbf{p}_i = \mathbf{0}. \tag{17}$$

We get the final $\mathbf{n}_i$ by summing all the solutions obtained by (17). Obviously, we have

$$\mathbf{m}_i \cdot \mathbf{p}_i \neq 0 \tag{18}$$

and

$$\mathbf{m}_i \cdot \mathbf{p}_j = 0, \quad i \neq j. \tag{19}$$

We acquire the diagonal matrix $\mathbf{Q}$ by

$$\mathbf{Q} = \mathbf{P}^T\mathbf{M}. \tag{20}$$

From (18) and (19), we obtain $Q_{ii} = \mathbf{m}_i \cdot \mathbf{p}_i \neq 0$ and $Q_{ij} = \mathbf{m}_i \cdot \mathbf{p}_j = 0$. The fact reflects that the diagonal matrix $\mathbf{Q}$ exists inverse matrix $\mathbf{Q}^{-1}$. Therefore, we transfer (13) into

$$\begin{aligned} \mathbf{P}^T\mathbf{Y} &= \mathbf{P}^T\mathbf{M}\mathbf{A} \\ &= \mathbf{Q}\mathbf{A}. \end{aligned} \tag{21}$$

Obviously, the abundance matrix can be easily computed by

$$\mathbf{A} = \mathbf{Q}^{-1}\mathbf{P}^T\mathbf{Y}. \tag{22}$$

Especially, when we do not impose the ANC and the ASC on the abundance matrix $\mathbf{A}$, we observe that all negative values appearing in $\mathbf{A}$ are really close to zero from experimental results. Considering that fact, first, we make that the abundance estimated by OPA meets the ANC constraint by setting all negative abundances to zero or its opposite value. We then divide each abundance vector by their sum to ensure that the abundance estimated by OPA meets the ASC constraint. We refer to the algorithm of estimating abundance as OPA and present its pseudocode in Algorithm 2. From Algorithm 2, we observe that the proposed OPA algorithm for abundance estimation is very efficient because it only involves matrix operations. It is worth noting that OPE and OPA are applicable to the LMM model, and OPE will present excellent performance in endmember extraction under the pure pixel assumption. However, OPA needs an accurate endmember extraction algorithm to extract endmember signatures, and then, it can obtain good abundance estimation results. Especially, in Section III, experimental results validate that our proposed OPA algorithm has low computational complexity in the task of abundance estimation on synthetic data and real Cuprite data.

III. Experimental Evaluation

We use both synthetic and real Cuprite data and conduct a series of experiments on them to evaluate the effectiveness and efficiency of our proposed OPE and OPA algorithms in the tasks of extracting endmembers and estimating the abundance of each endmember at every pixel. We perform the comparison tests using MATLAB 2018b on an i5-8300 CPU (2.3G). The endmember extraction capacity of our proposed OPE algorithm is assessed with five typical endmember extraction algorithms, i.e., OSP, VCA [12], ATGP [14], ADVMM, and SDVMM [15]. We use certain algorithms, e.g., hyperspectral subspace identification by minimum error (Hysime) [40], to filter the noise of HSI data in advance. In addition, we experimentally evaluate the performance of the proposed OPA algorithm and SUnSPI, CSUnL0, and typical FCLS algorithm [33] in the task of estimating abundance. Especially, considering that SUnSPI and CSUnL0 require the spectral library to complete the final abundance estimation,

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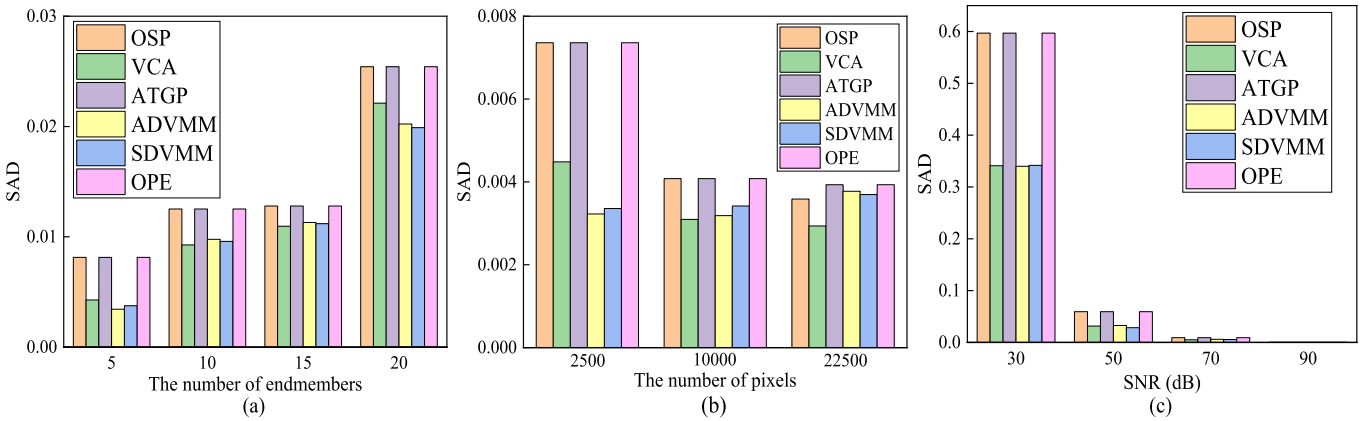


Fig. 2. SAD obtained by different endmember extraction algorithms on synthetic data (containing pure pixels) under (a) different number of endmembers, (b) different number of pixels, and (c) different levels of noise.

we only use them as comparison methods to evaluate the performance of our proposed OPA method on synthetic data.

Let $\hat{\mathbf{m}}_i$ denote the estimated endmember signature by OPE and $\mathbf{m}_i$ are the true endmember signature. We use the spectral angle distance (SAD) to evaluate the estimated endmembers with the ground-truth ones used in our experiments by

$$\text{SAD} = \sqrt{\frac{1}{c}\sum_{i=1}^{c}\left[\arccos\left(\frac{\hat{\mathbf{m}}_i^T\mathbf{m}_i}{\|\hat{\mathbf{m}}_i\|_2\|\mathbf{m}_i\|_2}\right)\right]^2} \tag{23}$$

where c is the number of endmembers. Let $\hat{\mathbf{A}}$ denote the estimated abundance matrix by OPA and $\mathbf{A}$ be the true abundance matrix. Simultaneously $\hat{a}_{ij}$ and a_{ij} are the ijth element of $\hat{\mathbf{A}}$ and $\mathbf{A}$, respectively. We use the root-mean-square error (RMSE) to evaluate the perforamnce of OPA in abundance estimation by

$$\text{RMSE} = \sqrt{\frac{1}{N}\frac{1}{c}\sum_{i=1}^{N}\sum_{j=1}^{c}(\hat{a}_{ij}-a_{ij})^2} \tag{24}$$

where N is the number of pixels of HSI dataset. Especially, OPE and OPA get the most accurate results in endmember extraction and abundance estimation when the values of SAD and RMSE are zero.

A. Synthetic Data

The synthetic data are created based on the USGS library. We generate synthetic data based on the linear mixture model [12], [39]. We consider two big families of synthetic data, i.e., with and without pure pixel assumption, to evaluate the effectiveness and efficiency of our proposed OPE and OPA algorithms in extracting endmembers and estimating the abundance of each endmember at every pixel. Each family of synthetic data has three types.

1) Synthetic data with 50 × 50 pixels and 70-dB noise under different numbers of endmembers, i.e., 5, 10, 15, and 20. We use this type of synthetic data to evaluate the influence of the number of endmembers on the performance of different algorithms in endmember extraction and abundance estimation.
2) Synthetic data with five endmembers and 70-dB noise under a different number of pixels, i.e., 2500, 10 000, and 22 500. The synthetic data are mainly used to exploit the impact of the number of pixels on the performance of different algorithms in the tasks of endmember extraction and abundance estimation.
3) Synthetic data with 50 × 50 pixels and five endmembers under different levels of noise, i.e., 30, 50, 70, and 90 dB. The synthetic data are designed to evaluate the robustness of different algorithms in the endmember extraction and abundance estimation.

1) Synthetic Data With Pure Pixel Assumption: We employ SAD to evaluate the performance of different algorithms in extracting endmember on three types of synthetic data. Fig. 2 shows the results of SAD obtained by different endmember extraction algorithms on different types of synthetic data. From Fig. 2(a), we see that the values of SAD of all algorithms will increase with the number of endmembers, but our proposed OPE algorithm keeps almost the same accuracy as other popular algorithms. From Fig. 2(b), we observe that the values of SAD will be smaller when the number of pixels increases, and our proposed OPE algorithm also has good performance in endmember extraction compared to other algorithms. From Fig. 2(c), we also observe that our proposed OPE algorithm has good robustness to noise. Therefore, our proposed OPE algorithm is a very effective endmember extraction algorithm.

We employ RMSE as the performance metric and use SUnSPI, CSUnL0, and FCLS as comparison methods to evaluate the performance of the proposed OPA method on different types of synthetic data in the task of abundance estimation, and the results of RMSE are presented in Tables I–III. From Table I, we see that OPA-based methods get more accurate results compared with SUnSPI and CSUnL0, and almost all OPA-based methods are better than the corresponding FCLS-based methods. From Table II, we observe that all OPA-based methods not only are better than SUnSPI, CSUnL0, and the corresponding FCLS-based methods but also obtain the most accurate results in terms of RMSE (its value is very close to zero). From Table III, we obtain that all OPA-based methods have better robustness compared to SUnSPI, CSUnL0, and the

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TABLE I

RMSE OBTAINED BY DIFFERENT ABUNDANCE ESTIMATION ALGORITHMS ON SYNTHETIC DATA (CONTAINING PURE PIXELS) WITH 2500 PIXELS AND 70-dB NOISE UNDER A DIFFERENT NUMBER OF ENDMEMBERS

The number of endmembers	SUnSPI	CSUnL0	OSP		VCA		ATGP		ADVMM		SDVMM		OPE	
			FCLS	OPA	FCLS	OPA	FCLS	OPA	FCLS	OPA	FCLS	OPA	FCLS	OPA
5	0.09094	0.01448	0.11106	**0.00001**	0.12256	**0.00001**	0.11106	**0.00001**	0.12390	**0.00001**	0.12313	**0.00002**	0.12365	**0.00003**
10	0.03504	0.01953	**0.04624**	0.09112	0.04105	**0.01059**	0.04624	**0.09112**	0.04898	**0.00946**	0.04614	**0.00198**	0.04619	**0.00248**
15	0.02111	0.01233	0.03348	**0.00532**	0.03339	**0.00222**	0.03348	**0.00532**	0.03362	**0.00363**	0.03463	**0.00051**	0.03233	**0.00332**
20	0.01774	0.01240	0.02994	**0.00124**	0.03001	**0.00421**	0.02994	**0.00124**	0.03059	**0.02630**	0.02918	**0.0044**2	0.03071	**0.00771**

TABLE II

RMSE OBTAINED BY DIFFERENT ABUNDANCE ESTIMATION ALGORITHMS ON SYNTHETIC DATA (CONTAINING PURE PIXELS) WITH FIVE ENDMEMBERS AND 70-dB NOISE UNDER A DIFFERENT NUMBER OF PIXELS

The number of pixles	SUnSPI	CSUnL0	OSP		VCA		ATGP		ADVMM		SDVMM		OPE	
			FCLS	OPA	FCLS	OPA	FCLS	OPA	FCLS	OPA	FCLS	OPA	FCLS	OPA
2500	0.11246	0.04831	0.12312	**0.00008**	0.09580	**0.00002**	0.12312	**0.00008**	0.09429	**0.00002**	0.12348	**0.00005**	0.09376	**0.00002**
10000	0.09034	0.00132	0.10883	**0**	0.12182	**0**	0.10883	**0**	0.07686	**0**	0.10886	**0**	0.12198	**0**
22500	0.07992	0.00238	0.07595	**0**	0.11978	**0**	0.10733	**0**	0.07696	**0**	0.11990	**0**	0.12015	**0**

TABLE III

RMSE OBTAINED BY DIFFERENT ABUNDANCE ESTIMATION ALGORITHMS ON SYNTHETIC DATA (CONTAINING PURE PIXELS) WITH FIVE ENDMEMBERS AND 2500 PIXELS UNDER DIFFERENT LEVELS OF NOISE

SNR (dB)	SUnSPI	CSUnL0	OSP		VCA		ATGP		ADVMM		SDVMM		OPE	
			FCLS	OPA	FCLS	OPA	FCLS	OPA	FCLS	OPA	FCLS	OPA	FCLS	OPA
30	0.11082	0.00341	0.11292	**0.00186**	0.11225	**0.00239**	0.11292	**0.00186**	0.11195	**0.00197**	0.11453	**0.00187**	0.11122	**0.00203**
50	0.09310	0.00689	0.10919	**0.00022**	0.09352	**0.00018**	0.10919	**0.00022**	0.10825	**0.00019**	0.07771	**0.00014**	0.10799	**0.00025**
70	0.08796	0.01683	0.12255	**0.00013**	0.12297	**0.00001**	0.12255	**0.00013**	0.12312	**0.00004**	0.11111	**0.00001**	0.09540	**0.00013**
90	0.07941	0.00652	0.08021	**0**	0.12193	**0**	0.08021	**0**	0.11140	**0**	0.10865	**0**	0.07659	**0**

corresponding FCLS-based methods. The above three results reflect that our proposed OPA algorithm is very effective in the task of abundance estimation.

In addition to evaluating the effectiveness of our proposed OPE and OPA algorithms, we also test their efficiency on three different types of synthetic data. Fig. 3 shows the processing time of different endmember extraction algorithms on different types of synthetic data. From Fig. 3(a), we see that OSP and ATGP need more time to complete endmember extraction with the increase of the number of endmembers, but VCA, ADVMM, and SDVMM keep a stable state. Most importantly, our proposed OPE algorithm requires less time on any number of endmembers compared to other algorithms. From Fig. 3(b), we observe that all algorithms both need more time when the number of pixels increases. Similarly, the efficiency of our proposed OPE algorithm is the highest compared to other algorithms. From Fig. 3(c), we obtain that almost all algorithms both maintain a relatively stable state. However, our proposed OPE algorithm is still the most efficient. Moreover, we consider the results of all endmember signatures extracted by different endmember extraction algorithms, i.e., VCA, ATGP, ADVMM, SDVMM, and OPE, to evaluate the efficiency of SunSPI, CSUnL0, FCLS, and our proposed OPA algorithms in the task of abundance estimation, and Fig. 4 shows their processing time on different types of synthetic data. From Fig. 4(a) and (b), we see that the FCLS-based algorithms (considering different endmember extraction results) need more time to complete abundance estimation with the growth of the number of endmembers and pixels, but OPA-based algorithms keep a stable state on any number of endmembers and pixels and require less time in the task of abundance estimation compared to FCLS-based algorithms. Moreover, SunSPI and CSUnL0 need more time compared with OPA-based algorithms and FCLS-based algorithms. From Fig. 4(c), we infer that all algorithms both maintain a relatively stable state on any noise levels and OPA-based algorithms are still the most efficient compared with SUnSPI, CSUnL0, and the corresponding FCLS-based algorithms. To sum up, our proposed OPE and OPA algorithms for endmember extraction and abundance estimation are not only very effective but also very efficient.

2) Synthetic Data Without Pure Pixel Assumption: To evaluate the performance of the proposed OPE algorithm in the task of endmember extraction on synthetic data without pure pixels, we employ SAD as the performance metric and conduct some related experiments on three different types of

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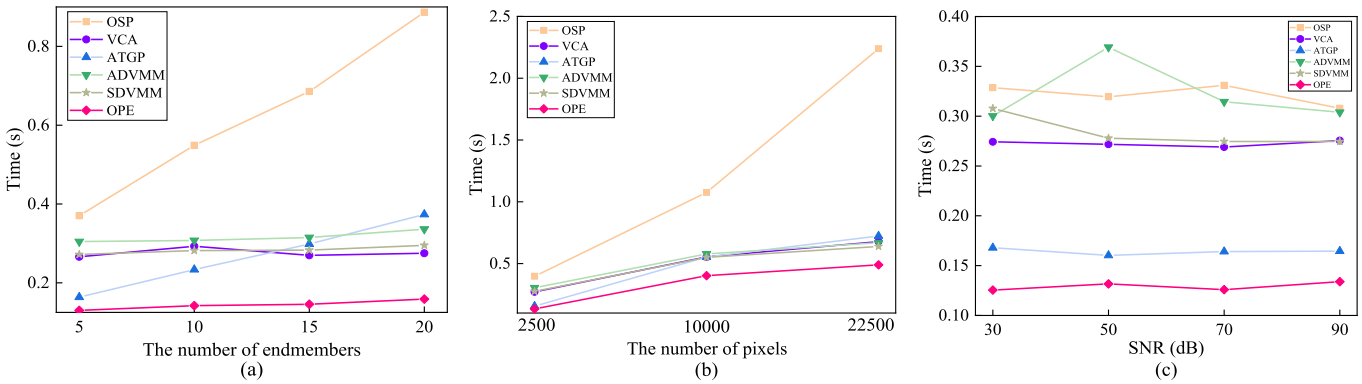


Fig. 3. Processing time (seconds) of different endmember extraction algorithms on synthetic data (containing pure pixels) under (a) different number of endmembers, (b) different number of pixels, and (c) different levels of noise.

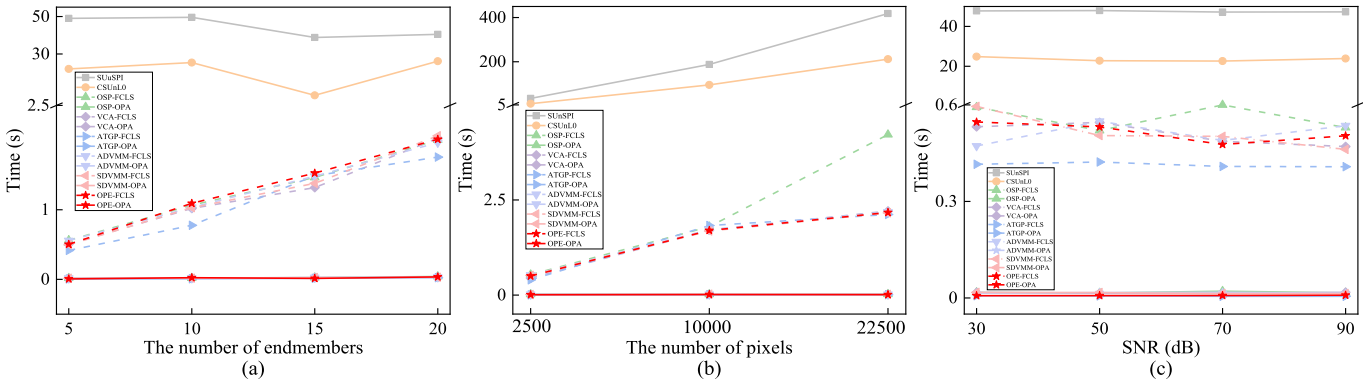


Fig. 4. Processing time (seconds) of different abundance estimation algorithms on synthetic data (containing pure pixels) under (a) different number of endmembers, (b) different number of pixels, and (c) different levels of noise.

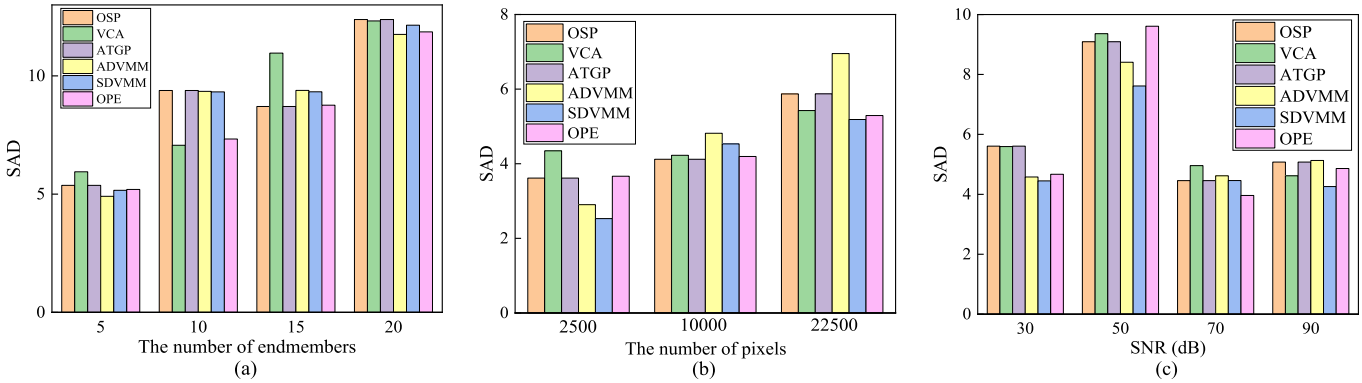


Fig. 5. SAD obtained by different endmember extraction algorithms on synthetic data (maximum purity of 0.8) under (a) different number of endmembers, (b) different number of pixels, and (c) different levels of noise.

synthetic data. An accurate endmember extraction algorithm has a small value of SAD. Fig. 5 shows the values of SAD obtained by different endmember extraction algorithms on three different types of synthetic data. From Fig. 5(a), we see that the increase of the number of endmembers will reduce the accuracy of all algorithms in endmember extraction. From Fig. 5(b), we observe that the performance of all endmember extraction algorithms basically maintains a stable state with the increment of the number of pixels. From Fig. 5(c), we observe that synthetic data with 50-dB noise will affect the value of SAD compared to synthetic data with other noise levels. The above three results show that our proposed OPE algorithm is not the best algorithm, but it maintains almost the same accuracy as other algorithms. Therefore, our proposed OPE algorithm is an effective endmember extraction algorithm.

We employ RMSE as the performance metrics to test the performance of SUnSPI, CSUnL0, FCLS, and OPA on three different types of synthetic data without pure pixels. The related results of RMSE are presented in Tables IV–VI. From Table IV, we see that all OPA-based algorithms almost

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TABLE IV

RMSE OBTAINED BY DIFFERENT ABUNDANCE ESTIMATION ALGORITHMS ON SYNTHETIC DATA (MAXIMUM PURITY OF 0.8) WITH 2500 PIXELS AND 70-dB NOISE UNDER A DIFFERENT NUMBER OF ENDMEMBERS

The number of endmembers	SUnSPI	CSUnL0	OSP		VCA		ATGP		ADVMM		SDVMM		OPE	
			FCLS	OPA	FCLS	OPA	FCLS	OPA	FCLS	OPA	FCLS	OPA	FCLS	OPA
5	0.05308	0	0.14065	**0.01061**	0.13302	**0.00426**	0.14065	**0.01061**	0.14743	**0.00320**	0.12222	**0.01122**	0.13292	**0.00413**
10	0.03093	0.02049	0.07898	**0.00528**	0.08014	**0.02595**	0.07898	**0.00528**	0.07942	**0.00396**	0.06853	**0.01613**	0.07377	**0.02014**
15	0.01691	0.01168	0.05155	**0.01907**	0.06014	**0.01689**	0.05155	**0.01907**	0.06014	**0.03853**	0.05040	**0.01177**	0.05249	**0.01564**
20	0.00974	0.00593	0.04541	**0.01956**	0.05001	**0.01201**	0.04541	**0.01956**	0.04751	**0.01093**	0.04504	**0.05630**	0.04941	**0.01562**

TABLE V

RMSE OBTAINED BY DIFFERENT ABUNDANCE ESTIMATION ALGORITHMS ON SYNTHETIC DATA (MAXIMUM PURITY OF 0.8) WITH FIVE ENDMEMBERS AND 70-dB NOISE UNDER A DIFFERENT NUMBER OF PIXELS

The number of pixels	SUnSPI	CSUnL0	OSP		VCA		ATGP		ADVMM		SDVMM		OPE	
			FCLS	OPA	FCLS	OPA	FCLS	OPA	FCLS	OPA	FCLS	OPA	FCLS	OPA
2500	0.10287	0.0001	0.15327	**0.00937**	0.13493	**0.01732**	0.15327	**0.00937**	0.14755	**0.00315**	0.15006	**0.00337**	0.13877	**0.01003**
10000	0.10127	0.05243	0.15072	**0.00470**	0.15440	**0.00416**	0.15072	**0.00470**	0.13351	**0.00350**	0.14776	**0.00429**	0.14815	**0.00347**
22500	0.09462	0.05262	0.14990	**0.00526**	0.12503	**0.00560**	0.11469	**0.00858**	0.13903	**0.00322**	0.11409	**0.00847**	0.15531	**0.00928**

TABLE VI

RMSE OBTAINED BY DIFFERENT ABUNDANCE ESTIMATION ALGORITHMS ON SYNTHETIC DATA (MAXIMUM PURITY OF 0.8) WITH FIVE ENDMEMBERS AND 2500 PIXELS UNDER DIFFERENT LEVELS OF NOISE

SNR (dB)	SUnSPI	CSUnL0	OSP		VCA		ATGP		ADVMM		SDVMM		OPE	
			FCLS	OPA	FCLS	OPA	FCLS	OPA	FCLS	OPA	FCLS	OPA	FCLS	OPA
30	0.08959	0.05321	0.19535	**0.10531**	0.19271	**0.06483**	0.19535	**0.10531**	0.15900	**0.12292**	0.14197	**0.21858**	0.16270	**0.04545**
50	0.07811	0.05501	0.14459	**0.00682**	0.14809	**0.02228**	0.14459	**0.00682**	0.13449	**0.00342**	0.09217	**0.00656**	0.15385	**0.01478**
70	0.10233	0.05154	0.12348	**0.00440**	0.10606	**0.02266**	0.12348	**0.0044**	0.12472	**0.00353**	0.09801	**0.00441**	0.11928	**0.00433**
90	0.07665	0.05411	0.13686	**0.00610**	0.16143	**0.00927**	0.13686	**0.0061**	0.11537	**0.00323**	0.13325	**0.00445**	0.14313	**0.00354**

get more accurate results compared with the corresponding FCLS-based algorithms. Moreover, SUnSPI is better than OPA-based algorithms on synthetic data with 20 endmembers, and CSUnL0 is better than OPA-based algorithms on synthetic data with 5 and 20 endmembers. From Table V, we obtain that OPA-based algorithms are better than the corresponding FCLS-based algorithms in the task of abundance estimation based on different endmember extraction results. Moreover, OPA-based algorithms are generally better than SUnSPI and CSUnL0. From Table VI, we observe that OPA-based algorithms have smaller RMSE values compared to SUnSPI, CSUnL0, and the corresponding FCLS-based algorithms, and they get more accurate results than SUnSPI, CSUnL0, and FCLS-based algorithms in abundance estimation. Moreover, although the advantages of OPA over FCLS are not obvious on synthetic data without pure pixels, OPA-based algorithms are still better than SUnSPI, CSUnL0, and the corresponding FCLS-based algorithms under any number of endmembers, any number of pixels, and any noise levels. Therefore, our proposed OPA algorithm is very effective in the task of estimating the abundance of each endmember at every pixel.

Fig. 6 shows the efficiency of different endmember extraction algorithms. In terms of endmember extraction, from Fig. 6(a), we see that OSP and ATGP require more time when the number of endmembers increases, ADVMM fluctuates up and down, and VCA, SDVMM, and our proposed OPE algorithm keep a stable state. Moreover, compared to other endmember extraction algorithms, our proposed OPE algorithm is the most efficient endmember extraction algorithm at any number of endmembers. From Fig. 6(b), we see that all endmember extraction algorithms both require more time with the growth of the number of pixels, but OPE still has the highest efficiency compared to other algorithms at any number of pixels. From Fig. 6(c), we obtain that all algorithms both maintain a stable state at any noise levels, but our proposed OPE algorithm is still the most efficient. In addition, we use all results of different endmember extraction algorithms to evaluate the efficiency of SUnSPI, CSUnL0, OPA-based algorithms, and FCLS-based algorithms in abundance estimation,

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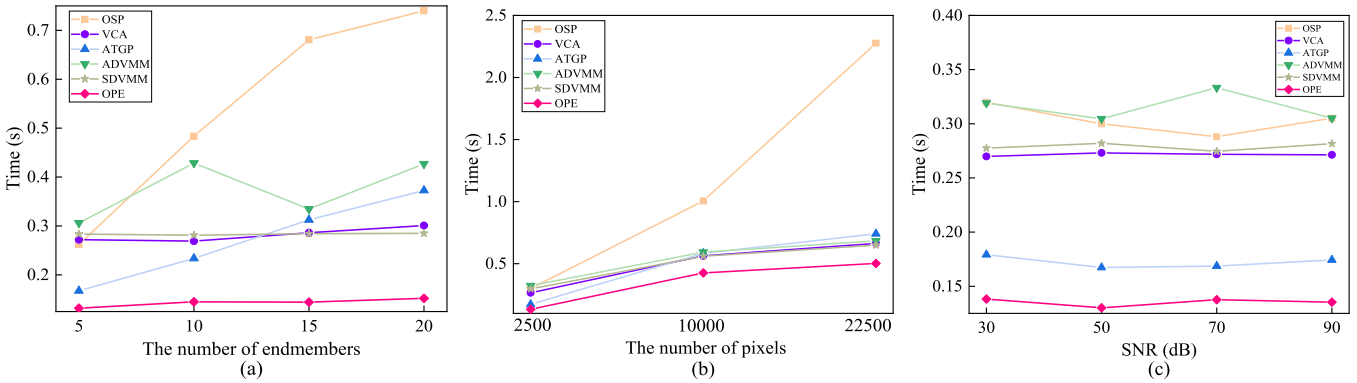


Fig. 6. Processing time (seconds) of different endmember extraction algorithms on synthetic data (maximum purity of 0.8) under (a) different number of endmembers, (b) different number of pixels, and (c) different levels of noise.

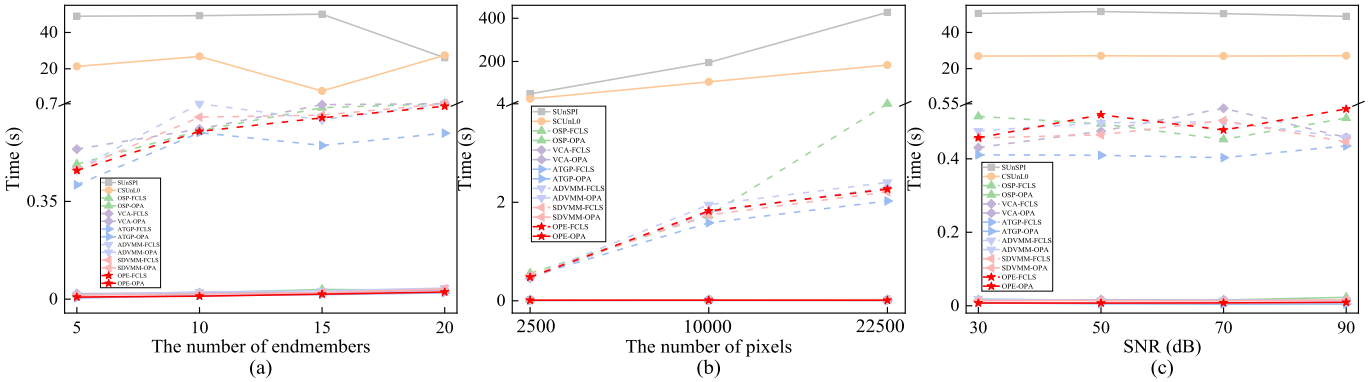


Fig. 7. Processing time (seconds) of different abundance estimation algorithms on synthetic data (maximum purity of 0.8) under (a) different number of endmembers, (b) different number of pixels, and (c) different levels of noise.

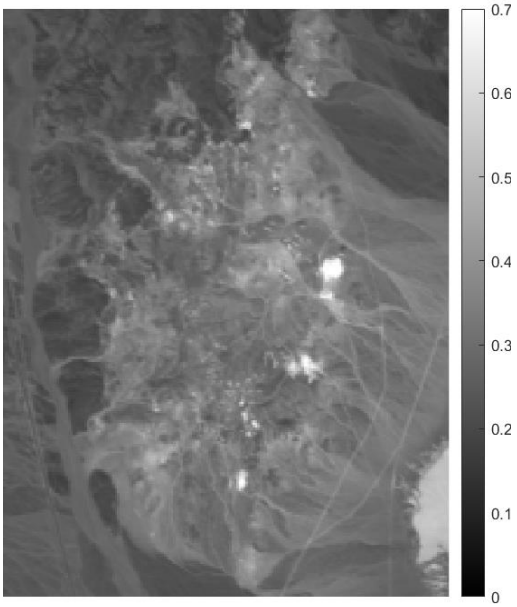


Fig. 8. 30th band of the considered AVIRIS Cuprite subimage.

and their processing time is presented in Fig. 7. From Fig. 7(a), we see that the processing time of SUnSPI, CSUnL0, and FCLS-based algorithms in abundance estimation fluctuates up and down with the growth of the number of endmembers. Moreover, we also see that OPA-based algorithms keep a stable state at any number of endmembers and require less time compared to SUnSPI, CSUnL0, and the corresponding FCLS-based algorithms. From Fig. 7(b), we observe that the processing time of SUnSPI, CSUnL0, and FCLS-based algorithms in abundance estimation will increase with the growth of the number of pixels. Simultaneously, OPA-based algorithms still keep a stable state and have the highest efficiency at any number of pixels compared to SUnSPI, CSUnL0, and the corresponding FCLS-based algorithms. From Fig. 7(c), we obtain that SUnSPI, CSUnL0, FCLS-based algorithms, and OPA-based algorithms have a relatively stable state, but OPA-based algorithms have the lowest processing time at any noise levels in abundance estimation. To sum up, our proposed OPE and OPA algorithms for endmember extraction and abundance estimation are not only very effective but also very efficient.

B. Real Cuprite Data

We use the well-known AVIRIS Cuprite data[2] to evaluate the effectiveness of efficiency of our proposed OPE and OPA algorithms in the endmember extraction and abundance estimation. The portion used in experiments corresponds to

[2]http://aviris.jpl.nasa.gov/html/aviris.freedata.html

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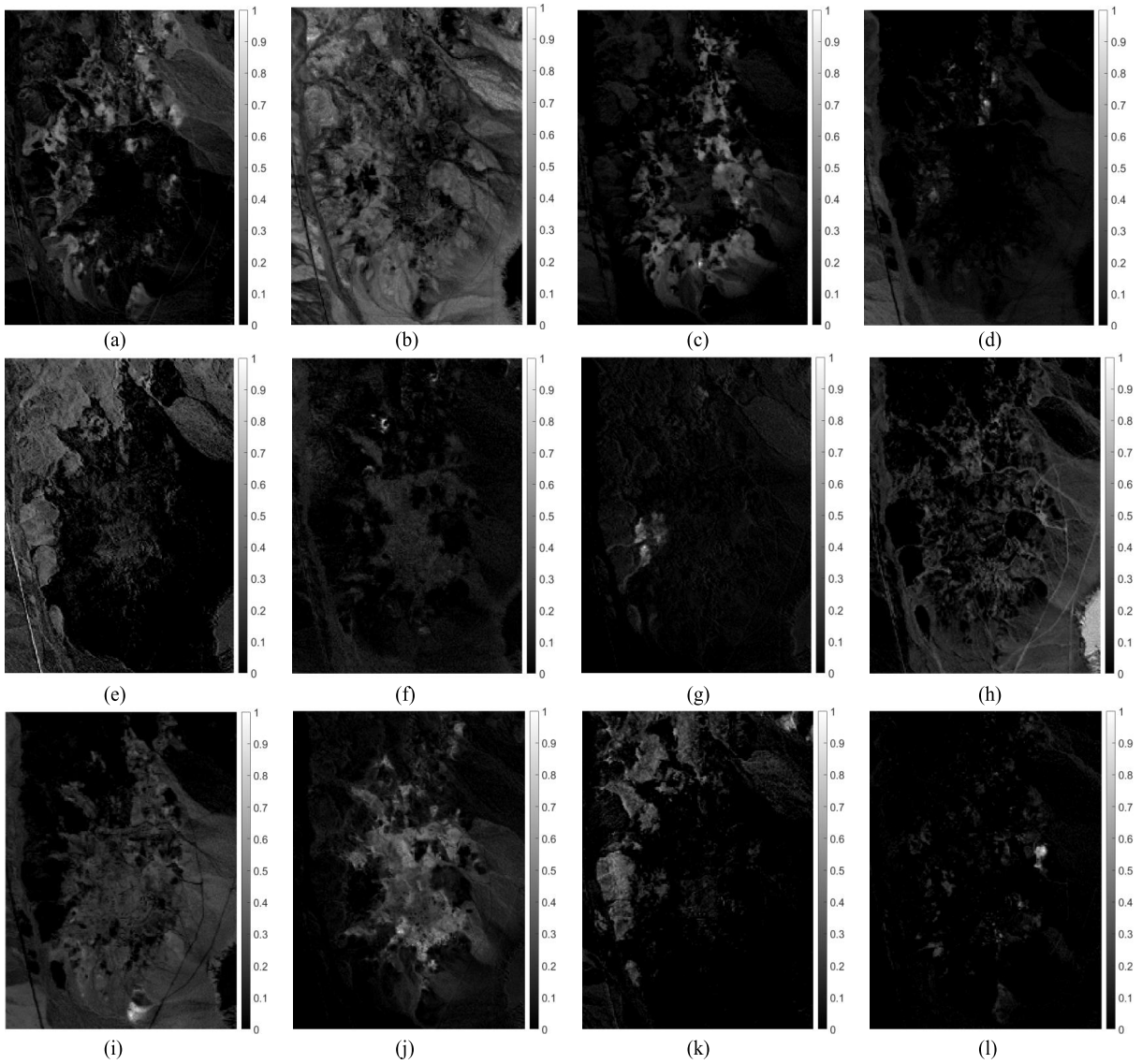


Fig. 9. Abundance fraction maps estimated by our proposed OPA algorithm from the Cuprite subimage. (a) Montmorillonite SCa-2.b. (b) Jarosite WS368 Pb #1. (c) Montmorillonite CM20. (d) Nontronite SWa-1.a. (e) Montmorillonite + Illi CM37 #1. (f) Nontronite NG-1.a. (g) Kaolin/Smect H89-FR-5 30K #1. (h) Montmorillonite + Illi CM37 #2. (i) Kaolin/Smect H89-FR-5 30K #2. (j) Kaolin/Smect H89-FR-5 30K #3. (k) Jarosite WS368 Pb #2. (l) Chalcedony CU91-6A.

TABLE VII
SAD AND THE PROCESSING TIME OBTAINED BY DIFFERENT ENDMEMBER EXTRACTION ALGORITHMS ON CUPRITE SUBIMAGE

Metrics	OSP	VCA	ATGP	ADVMM	SDVMM	OPE
SAD	4.3723	4.8400	4.3723	4.6830	**4.3461**	4.4785
Time	5.9829	**3.6465**	4.3723	4.07224	3.9454	4.0650

a 250 × 191 pixel subset of Cuprite data, with 224 spectral bands in the range 0.4–2.5 μm and a nominal spectral resolution of 10 nm. Due to water absorption and low SNR, we remove bands 1–6, 105–115, 150–170, and 222–224 and retain 183 spectral bands in our experiments. Fig. 8 shows the 30th band of Cuprite subimage for this experiment.

We employ SAD to evaluate the effectiveness of our proposed OPE algorithm on the Cuprite subimage. Table VII shows the values of SAD obtained by different endmember extraction algorithms and their processing time. From Table VII, we see that our proposed OPE endmember extraction algorithm is not optimal or the fastest, but it gets almost the same results as other algorithms. Therefore, our proposed OPE algorithm is an effective endmember extraction algorithm.

Moreover, Fig. 9 shows the abundance maps obtained by our proposed OPA algorithm. Simultaneously, we also consider the reconstruction error (RSE) to evaluate the performance of our proposed OPA algorithm in abundance estimation by

$$\text{RSE} = \frac{||\mathbf{Y} - \hat{\mathbf{Y}}||_F}{||\mathbf{Y}||_2} \tag{25}$$

where $||\cdot||_F$ indicates the Frobenius norm, $||\cdot||_2$ represents the L_2-norm, $\mathbf{Y}$ is the real Cuprite subimage dataset, and $\hat{\mathbf{Y}}$ is the estimated Cuprite subimage dataset. The related results of

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TABLE VIII
RSE AND THE PROCESSING TIME OF DIFFERENT ABUNDANCE ESTIMATION ALGORITHMS ON CUPRITE SUBIMAGE

Metrics	OSP		VCA		ATGP		ADVMM		SDVMM		OPE	
	FCLS	OPA	FCLS	OPA	FCLS	OPA	FCLS	OPA	FCLS	OPA	FCLS	OPA
RSE	**0.50515**	0.52086	**0.57674**	0.60858	**0.50515**	0.52086	**0.52792**	0.54824	**0.50978**	0.52063	**0.51160**	0.54986
Time	11.37464	**0.03294**	10.11134	**0.02912**	10.21231	**0.02803**	10.47131	**0.03055**	10.11796	**0.02843**	10.05766	**0.02345**

RSE and the processing time of FCLS and OPA in abundance estimation based on different endmember extraction algorithms are presented in Table VIII. From Table VIII, we see that OPA-based algorithms may not be optimal in abundance estimation, but they get almost the same results as FCLS-based algorithms. Simultaneously, OPA-based algorithms are always faster than FCLS-based algorithms in abundance estimation on the Cuprite subimage, and they are about 500 times faster than FCLS-based algorithms. The results obtained by OPA on synthetic data and real Cuprite data show that our proposed OPA algorithm can be considered as a new and fast abundance estimation algorithm to replace FCLS.

IV. CONCLUSION AND FUTURE WORK

We have developed a new and fast orthogonal projection framework to complete the overall hyperspectral unmixing task, i.e., extracting endmembers and estimating the abundance of each endmember at every pixel. We refer to the proposed algorithms for endmember extraction and abundance estimation as OPE and OPA, respectively. The proposed OPE algorithm completes endmember extraction by searching for the orthogonal vector of endmembers that have been extracted and chooses the pixels with the maximum projection to the orthogonal vector as endmembers. The proposed OPA algorithm computes a diagonal matrix whose multiplication with the endmember matrix is not only a square matrix but also a diagonal matrix and then exploits some specific operations of matrices to estimate the abundance of each endmember at every pixel. Both OPE and OPA only require fast matrix operations, and they have high efficiency in the tasks of endmember extraction and abundance estimation, respectively. Simultaneously, the proposed OPE and OPA algorithms have been evaluated on synthetic data and real Cuprite data, and the experimental results show that they are not only very efficient but also very effective in the endmember extraction and abundance estimation.

As with any new approach, there are some unresolved issues that may present challenges over time. Especially, we assume that HSI is represented by the multiplication of the endmember matrix and the abundance matrix. In our future work, we will investigate how to extract endmembers and estimate abundances from HSIs that are nonlinearly mixed.

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